Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Physical AI–Enabled XR and Spatial Digital Twin Platform for Multi-User Skill Training			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical - Materials		*Mechanical - Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Robot Vision and Production Automation Technologies*	*Industrial/General Machinery*	*Industrial/General Machinery Software*
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The proposed technology is a Physical AI–enabled XR training platform that transforms real-world environments and task procedures into interactive, multi-user training experiences. It combines a spatial digital twin of the training area with Physical AI capabilities—including real-time pose estimation, action/gesture recognition, and scene understanding—to guide, monitor, and evaluate trainees during hands-on activities. Core functions include scenario authoring for step-by-step workflows, real-time visual guidance (XR overlays, checklists, safety cues), automated step verification, and adaptive coaching to correct errors as they occur. The platform records trainee actions, supports session replay, and generates automated performance reports (time, accuracy, sequence compliance, safety-critical misses). Potential uses include industrial safety and compliance training, manufacturing and logistics operations, emergency response drills, defense and public-sector readiness, and education-based skill training. The modular design supports rapid scenario updates and scalable deployment across diverse industries and facilities.			

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Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* Modern workplaces across manufacturing, logistics, public safety, and defense require personnel to master complex physical procedures faster, with higher safety and measurable competency. Digital-twin–based training is increasingly recognized as a way to shift from static instruction to evidence-driven, competency-traceable learning with adaptive content delivery and improved traceability of trainee performance [1–3]. In parallel, XR/AR training has demonstrated strong potential to improve hands-on learning by enabling immersive practice, step-by-step guidance, and reduced dependence on scarce equipment or expert instructors, and these benefits extend well beyond aviation to broader industrial workforce training [4–8]. However, scalable deployment still faces persistent barriers such as robust spatial tracking/registration, workflow integration, user workload, and reliable evaluation, which typically require iterative validation in real environments beyond lab prototypes [9–11]. Recent studies also highlight the value of Physical AI—using pose/action recognition and real-time feedback—to improve movement quality and skill acquisition, reinforcing the need for AI-driven coaching inside training systems [12]. As a Professor of Aerospace Engineering, Dr. Joon Chung has led numerous high-profile collaborative projects at the interface of academia and industry: • Holodeck Project (Bombardier-led): Developed AR-enabled Electronic Flight Instrument Systems (EFIS) for real-time cockpit interface prototyping and evaluation. • AI-enabled Metaverse Campus (OVA-led, Canada's Digital Supercluster): Introduced haptic gloves and immersive training checklists for Canadair CL-515 operations. • Advanced Airspace Usability (ADAIR, Bombardier-led consortium): Development of conformal HUD symbologies for trajectory-based operations, addressing cockpit decision support. • Mitacs–Columbiad Projects (multiple IT-series grants): XR-enabled control stations with multi-user collaboration, VR, and haptics for remote operation of UAVs, and space applications. This joint R&D is needed to bridge the gap between promising prototypes and a validated, deployable Physical AI training platform. The research institute will contribute core innovation in Physical AI (pose/action recognition, scene understanding, step verification, adaptive coaching), spatial digital-twin modeling, and synthetic-data pipelines for scalable AI training and verification [12]. The Korean SME partner will provide real operational requirements, constraints, and productization pathways, including integration into training delivery workflows and access to representative users and environments for validation. Collaboration is essential because Physical AI performance depends on real-world variability (occlusions, multi-user motion, and space layouts), and high-accuracy alignment remains a known challenge in AR-assisted tasks [13, 14]. Together, the partners will deliver a TRL-5 validated system that reduces critical errors, accelerates skill acquisition, and enables auditable competency assessment across multiple industries.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Physical AI (Pose/Action Recognition) 2. Extended Reality (XR) Training 3. Spatial Digital Twin 4. Scenario-Based Skill Training 5. Automated Performance Analytics

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* his R&D project will develop a Physical AI–enabled XR and digital-twin training technology suite for complex, safety-critical tasks across multiple industries. We will develop: (1) a spatial digital-twin content pipeline that converts real facilities and task-relevant engineering artifacts (e.g., layouts, CAD models, SOPs, and operational constraints) into interactive XR training environments; (2) a scenario authoring, execution, and replay engine that represents procedures as step-by-step workflows with hazards, pass/fail criteria, and full event logging for traceable competency assessment; (3) Physical AI modules—including real-time pose estimation, action/gesture recognition, and scene understanding—to verify step completion, detect unsafe motions, and provide adaptive feedback during hands-on training; (4) real-time XR guidance tools such as contextual overlays, hands-free checklists, safety-zone visualization, and instructor annotations to support learning in realistic environments; (5) multi-user training modes enabling instructor–trainee collaboration, team-based drills, and synchronized shared-state practice for operational readiness; and (6) automated performance evaluation and analytics, including scoring of time, accuracy, sequence compliance, and safety-critical errors, with dashboards and exportable training reports. The technology will be modular and platform-agnostic to support integration with industrial training ecosystems and rapid updates to procedures and content. The project will define common APIs, asset formats, and validation protocols, and will deliver a TRL-5 prototype demonstrated in a relevant training environment within a three-year timeline.			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* Target specifications focus on real-time performance, Physical AI reliability, training coverage, multi-user collaboration, and measurable learning outcomes for safety-critical, procedure-based tasks across industries. 1) XR Runtime Performance and User Experience • Sustained real-time rendering at 72–90 FPS on target XR devices. • Motion-to-photon latency <20 ms to reduce discomfort and enable precise interaction. • Session stability with >99% crash-free runtime over standard training durations (20–40 minutes). 2) Spatial Digital Twin and Alignment Accuracy • Spatial mapping/anchoring accuracy enabling XR overlay alignment error ≤10 mm in representative training zones. • Orientation error ≤2° for key interaction elements (tools, control panels, safety boundaries). • Environment initialization / calibration time ≤10 minutes for a new training space, supporting rapid deployment. 3) Physical AI (Action Understanding) Performance • Pose estimation tracking reliability ≥95% under normal lighting and typical occlusion conditions. • Action/step recognition accuracy ≥90% for defined procedural actions (e.g., correct motion pattern, correct sequence). • Unsafe action detection sensitivity ≥90% for pre-defined critical violations (e.g., entering restricted zones, missing safety checks). • Real-time feedback latency ≤200 ms from action detection to user feedback. 4) Scenario Coverage and Traceability • Minimum 30 training scenarios total, including at least 20 standard operating procedures and 10 abnormal/safety-critical variations. • Step-level logging coverage ≥95% of defined actions, including time stamps, sequencing,			

	and error flags. • Replay and audit export available for 100% of completed sessions. 5) Multi-user Collaboration and Synchronization • End-to-end collaboration latency ≤200 ms for instructor–trainee interaction. • Shared-state synchronization consistency ≥99% across multi-user sessions. • Support for 2–6 concurrent users in one training space depending on scenario type. 6) Automated Evaluation and Training Effectiveness • Event-capture reliability ≥95% for step completion, omissions, and out-of-order actions. • Automated report generation in <60 seconds per session. • Demonstrated training improvement vs. baseline methods: ≥25% reduction in task completion time and ≥30% reduction in critical errors, verified through controlled evaluations with standardized protocols.
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* The R&D plan is structured as a three-year, TRL 2→5 program progressing from concept definition and proof-of-principle to a validated prototype demonstrated in a relevant training environment. The approach integrates spatial digital twin engineering, XR interaction design, Physical AI (pose/action recognition + scene understanding), and rigorous validation through iterative user studies and controlled training assessments. **Year 1 – Foundations and Proof-of-Principle (TRL 2→3)** **Research approach:** Establish a robust technical foundation that can generalize across industries (manufacturing, logistics, safety training, public-sector readiness), while selecting representative **physical task categories** (step-based procedures, safety compliance tasks, team coordination drills). **Key tasks & methodologies** 1. **Requirements and use-case definition:** Collect training workflow requirements (task complexity, safety constraints, assessment needs, multi-user modes). Define success metrics and baseline comparison protocols. 2. **System architecture & interfaces:** Define modular architecture covering (i) spatial digital twin, (ii) scenario engine, (iii) Physical AI pipeline, (iv) multi-user orchestration, and (v) analytics/reporting. Establish data standards and APIs for scenario assets and training logs. 3. **Spatial digital twin v1:** Build a pipeline to represent training environments using simplified facility models or layouts, including anchors, safety zones, and interaction regions. 4. **Scenario representation v1:** Implement a scenario authoring framework using step graphs (actions, constraints, hazards, pass/fail criteria). Include event logging for traceability and replay. 5. **Physical AI baseline models:** Prototype pose tracking and simple action recognition for core motions (reach, pick/place, correct stance, zone entry/exit), focusing on real-time feasibility and robustness. **Expected milestones** • **M1.1** Requirements/specification baseline + evaluation metrics • **M1.2** Spatial digital twin pipeline v1 (environment + anchors + zones) • **M1.3** Scenario engine v1 (authoring + execution + replay + logs) • **M1.4** Proof-of-principle demo with **3–5** scenarios and basic scoring dashboard **Year 2 – Functional Prototype with Physical AI and Multi-User Training (TRL 3→4)** **Research approach:** Expand capability from single-user demonstration to functional training workflows with **real-time coaching**, improved fidelity, and collaborative training sessions. **Key tasks & methodologies** 1. **XR guidance & interaction expansion:** Add contextual overlays, hands-free checklists, safety cues, object/zone highlighting, and instructor annotations. Optimize UI/UX to reduce cognitive load and support rapid comprehension. 2. **Physical AI step verification:** Enhance action recognition to verify step completion, detect

out-of-order actions, and identify safety-critical violations. Introduce confidence scoring and human-in-the-loop calibration tools.

3. **Synthetic data pipeline:** Generate synthetic training samples from simplified 3D task models and environment layouts to improve recognition robustness under occlusion, lighting variation, and viewpoint changes.
4. **Multi-user instructor–trainee collaboration:** Implement synchronized session state, role permissions (instructor vs trainee), shared pointers/annotations, and team-based drills (2–6 concurrent users depending on scenario type).
5. **Training analytics v1:** Develop automated scoring logic (time, accuracy, sequence compliance, missed steps, safety violations), plus exportable reports and session summaries.

Validation activities

- Iterative usability testing (task completion friction, error types, learnability)
- Reliability testing (logging completeness, AI stability, multi-user sync)
- Mid-year pilot evaluations with representative users to refine scenario design and feedback timing

Expected milestones

- **M2.1** XR guidance toolkit v1 (overlays, checklists, safety cues)
- **M2.2** Physical AI verification v1 (pose/action + step checking)
- **M2.3** Multi-user training mode v1 (instructor–trainee + team sessions)
- **M2.4** Functional prototype demo covering **15–20** scenarios + usability validation report

Year 3 – Integration Hardening, Validation, and TRL-5 Demonstration (TRL 4→5)

Research approach: Mature the system into a stable prototype suitable for repeatable training delivery with measurable performance improvements and deployment readiness.

Key tasks & methodologies

1. **Performance engineering & robustness:** Optimize runtime performance (FPS/latency), stabilize AI inference for real-time feedback, and harden multi-user synchronization. Validate crash-free operation and consistent scoring across sessions.
2. **Deployment packaging & interoperability:** Finalize modular APIs, asset formats, scenario versioning, and training log exports to enable integration with external training operations and reporting workflows.
3. **Expanded scenario library:** Finalize a library of **≥30 scenarios**, including **≥20 standard procedures** and **≥10 abnormal/safety-critical variations**, ensuring consistent design patterns and assessment logic.
4. **Formal evaluation protocol:** Establish standardized metrics: time-to-complete, step compliance, safety violations, performance consistency, retention checks, and instructor workload. Conduct controlled studies comparing XR+Physical AI training versus baseline methods (paper/video instruction or conventional guided practice).
5. **TRL-5 demonstration:** Execute demonstrations in a relevant training setting using representative tasks, realistic constraints (space, time pressure, multi-user coordination), and repeatable assessment.

Expected milestones

- **M3.1** Integrated beta release + API/specification package + deployment guide
- **M3.2** Full scenario library (≥30 scenarios) + standardized scoring templates
- **M3.3** Controlled evaluation results demonstrating measurable performance gains
- **M3.4** TRL-5 demonstration + final technical report + commercialization roadmap

Project management rhythm (continuous across 3 years)

- Biweekly technical sprints and monthly milestone reviews
- Shared risk register (AI robustness, alignment accuracy, user workload, multi-user performance)
- Continuous documentation, version control, and reproducible evaluation datasets

This plan ensures progressive maturation of the technology from early concept to a field-validated **Physical AI XR training prototype**, with clear milestones, measurable targets, and a credible path

	to deployment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* The Research Institute, University of Toronto (U of T), will lead the core research, prototyping, and validation activities required to develop the Physical AI–enabled XR and spatial digital-twin training platform for complex, safety-critical tasks across industries. Technically, the institute will define the overall system architecture and technical requirements, establish the spatial digital-twin content pipeline (facility layouts/3D assets/SOPs → interactive XR environments), and develop the scenario representation framework (step graphs, constraints, hazards, pass/fail criteria, and full event logging with replay). The institute will design and implement XR interaction methods, including real-time visual guidance (contextual overlays, hands-free checklists, safety-zone cues), and multi-user instructor–trainee collaboration features (shared-state synchronization, annotation, and session management). Within the Physical AI scope, the institute will develop real-time pose estimation, action/gesture recognition, and scene understanding modules to enable step verification, unsafe-motion detection, and adaptive feedback during training. The institute will also build a synthetic-data generation workflow to improve model robustness, reduce dependence on costly real-world data capture, and support scalable expansion to new tasks and environments. In addition, the institute will implement automated evaluation pipelines including scoring logic, analytics dashboards, and exportable reports for competency traceability. The institute will define quantitative validation protocols covering runtime performance, AI reliability, usability, and training effectiveness (time-to-complete, sequence compliance, critical error reduction, and retention where applicable). Human factors and training science methodologies (task analysis, iterative usability testing, controlled comparisons to baseline training) will be conducted to demonstrate learning outcomes and support TRL progression. Resources provided by the institute include the XR simulation toolchain (Unreal Engine-based development), spatial digital-twin and scenario-authoring software modules, compute infrastructure, and XR hardware/test facilities required for iterative development and demonstrations. The institute will provide key research personnel (faculty leadership, postdoctoral researchers, graduate students, and technical staff) to deliver development, experimentation, documentation, and technical reporting. The institute will manage the R&D workplan, maintain version control and technical documentation, and support efficient knowledge transfer through regular technical reviews, joint sprints, and integration support to ensure the final prototype is validated, well-documented, and ready for partner-led deployment and commercialization. In addition to technical leadership, U of T offers a strong cost and commercialization advantage. Compared with typical R&D costs in the USA and Europe, Canada can provide approximately one-third lower cost for hiring graduate-level researchers, enabling deeper R&D execution within the same budget. U of T also provides industry-friendly IP practices and a lower overhead model (≈40% of direct costs), improving budget efficiency and accelerating translation to commercial prototypes. Finally, U of T will support IP strategy and dissemination through technical reports, TRL evidence packages, evaluation datasets (where permissible), and deployment guidelines aligned with partner productization objectives.
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* The Korean SME partner will serve as the primary industrial co-developer and downstream deployment pathway for the R&D outputs, ensuring that the Physical AI–enabled XR and spatial digital-twin training platform aligns with real operational training needs, product requirements, and Korean market adoption pathways. The SME will lead industry requirements definition and prioritization by providing representative use-cases, procedure structures, competency standards, safety constraints, and acceptance criteria for complex, safety-critical physical tasks across targeted

	sectors. The SME will also define integration requirements for industrial training ecosystems, including user management, training workflow orchestration, content packaging/versioning, interoperability specifications (APIs, data formats), and reporting needs for competency tracking. On the development side, the SME will implement deployment-oriented modules required to operationalize the prototype, including training session management, scenario distribution and update mechanisms, instructor/trainee role configuration, and production-grade user interfaces. The SME will contribute engineering effort toward performance optimization and reliability hardening for repeated training use, including deployment packaging, device compatibility testing, and operational support considerations. Where applicable, the SME will support localization and adaptation to Korea-relevant operational contexts by coordinating subject-matter experts, ensuring appropriate terminology and standards alignment, and assisting with content validation. For testing and validation, the SME will host and coordinate field-representative evaluations by recruiting target users (trainees, instructors, operators), providing access to suitable training spaces and representative task setups, and supporting iterative pilots to capture real-world variability (space layouts, occlusions, multi-user movement, safety constraints). The SME will co-define validation protocols and success metrics, participate in usability and acceptance testing, and assess scalability, maintainability, and integration readiness. Throughout the project, the SME will commit engineering staff time for co-development sprints and joint technical reviews, provide industrial project management and coordination, and lead commercialization planning activities such as customer discovery, market validation, and identification of early adopters in Korea. Close collaboration with the research institute will ensure effective knowledge transfer, clear interface ownership, and a credible pathway from TRL-validated prototype to deployable commercial training solutions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc. This joint R&D project is expected to deliver a validated Physical AI–enabled XR and spatial digital-twin training platform that measurably improves training effectiveness for complex, safety-critical tasks while reducing training cost and time. Technologically, the project will produce: (1) a modular spatial digital-twin pipeline that converts real facilities, layouts, and task documentation (e.g., SOPs, engineering assets) into interactive XR training environments; (2) a scenario authoring, execution, and replay engine that models procedures as step-based workflows with hazards, pass/fail criteria, and full event logging for competency traceability; (3) real-time XR guidance functions including contextual overlays, hands-free checklists, and safety-zone cues; (4) Physical AI modules (pose/action recognition and scene understanding) to verify step completion, detect unsafe motions, and deliver adaptive coaching; (5) multi-user instructor–trainee and team-training modes enabling collaborative practice in shared spaces; and (6) automated performance evaluation and analytics with reliable scoring and exportable training reports. By the end of the project, these capabilities will be demonstrated as a TRL-5 prototype in a relevant training environment with quantitative evidence of improved trainee performance (e.g., reduced critical errors and faster task completion compared to baseline methods). From a commercialization perspective, the outcomes create a strong pathway to productization as a scalable training solution for industrial workforce development, safety compliance, logistics and operations, public-sector readiness, and defense/education training programs. The modular design enables rapid creation and updating of scenario libraries for different facilities and procedures, supporting recurring revenue through licensing, content packages, and training analytics services. Synthetic-data–supported Physical AI reduces reliance on costly real-world data collection, accelerating deployment and improving maintainability as new scenarios are added. Broader impact includes strengthening Korea–Canada industrial collaboration in next-generation training technologies, improving workforce readiness, and enhancing safety and operational reliability through better procedural compliance and measurable competency assessment. Economically, the project can support high-skill jobs in Physical AI, XR software, digital-twin engineering, and training

	content development, while improving productivity by reducing rework, incidents, downtime, and training bottlenecks. The resulting platform also establishes a reusable foundation adaptable across multiple industrial sectors requiring hands-on skill training.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization